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Studierichting: Informatica
Oefeningenreeks 6E nr. 1.3

$T_5(f,0)(x)$ voor $f(x) = e^{\sin x}$

$$f(x) = e^{\sin x}$$

$$f'(x) = e^{\sin x} \cos x$$

$$f''(x) = e^{\sin x} \cos^2 x - e^{\sin x} \sin x$$

We gebruiken de bekende Taylorontwikkelingen rond 0:

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120}$$

$$e^u = 1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \frac{u^4}{4!} + \frac{u^5}{5!}$$

Met:

$$u = \sin x = x - \frac{x^3}{6} + \frac{x^5}{120}$$

Dus:

$$e^{\sin x} = 1 + \left(x - \frac{x^3}{6} + \frac{x^5}{120}\right) + \frac{\left(x - \frac{x^3}{6} + \frac{x^5}{120}\right)^2}{2!} + \frac{\left(x - \frac{x^3}{6} + \frac{x^5}{120}\right)^3}{3!} + \frac{\left(x - \frac{x^3}{6} + \frac{x^5}{120}\right)^4}{4!} + \frac{\left(x - \frac{x^3}{6} + \frac{x^5}{120}\right)^5}{5!}$$

Tot en met graad 5 geeft dit:

$$e^{\sin x} = 1 + x + \frac{x^2}{2} - \frac{x^4}{8} - \frac{x^5}{15}$$

Dus:

$$T_5(f,0)(x) = 1 + x + \frac{x^2}{2} - \frac{x^4}{8} - \frac{x^5}{15}$$

References